

SKILL_WEEK - 1

(SESSION NUMBER - 1)

WSL INSTALLATION

WSL Installation – mac-OS Explanation:

WSL (Windows Subsystem for Linux) is a feature specifically designed for the Windows operating system to provide a Linux environment without requiring a separate virtual machine. Since I am using mac-OS, WSL installation is not required. mac-OS is based on a Unix foundation and already provides a built-in Terminal with a Unix command-line environment. Therefore, OS practical commands such as `uname`, `ls`, `cd`, `mkdir`, `ps`, `top`, `df`, `gcc/clang`, and other Unix commands can be executed directly in the mac-OS Terminal without installing WSL.

Practical Evidence:

Screenshot 1: Checking the operating system using `uname`

1. `uname`

```
amrutha@Amruthas-MacBook-Air ~ % uname
Darwin
amrutha@Amruthas-MacBook-Air ~ %
```

Screenshot 2: Displaying detailed system information

2. `uname -a`

```
amrutha@Amruthas-MacBook-Air ~ % uname -a
Darwin Amruthas-MacBook-Air.local 25.5.0 Darwin Kernel Version 25.5.0: Tue Jun  9 22:26:22 PDT 2026; root:xnu-12377.121.10~1/RELEASE_ARM64_T8132 arm64
```

Screenshot 3: Checking CPU/process information

3. `Ps`

```
amrutha@Amruthas-MacBook-Air ~ % ps
  PID TTY          TIME CMD
 70833 ttys007      0:00.03 -zsh
amrutha@Amruthas-MacBook-Air ~ %
```

Screenshot 4: Checking system resource usage

4. top

```
Processes: 715 total, 4 running, 711 sleeping, 4176 threads 21:04:56
Load Avg: 3.07, 2.90, 3.43 CPU usage: 16.77% user, 7.65% sys, 75.56% idle
SharedLibs: 622M resident, 113M data, 87M linkedit. MemRegions: 0 total, 0B resident, 0B private, 1563M shared.
PhysMem: 15G used (2249M wired, 4644M compressor), 281M unused.
VM: 327T vsize, 6144M framework vsize, 4487(0) swapins, 75960(0) swapouts.
Networks: packets: 212913648/94G in, 21952808/15G out. Disks: 35663940/799G read, 20486552/368G written.

PID    COMMAND    %CPU    TIME    #TH    #WQ    #PORTS    MEM    PURG    CMPRS    PGRP    PPID    STATE    BOOSTS    %CPU_ME
32277  promecefplug 26.4    10:57:05 14    5    258    243M- 2288K+ 48M    32270 32270 sleeping *1[5]    0.03407
32270  wpsoffice    24.6    09:34:10 84/1  3    1823    1468M 208K    1014M 32270 1    running  *1[16416]    4.60005
48745  Google Chrom 21.7    50:46:38 43    1    620    774M+ 0B      383M    96762 96762 sleeping *0[7]    0.00000
587    WindowServer 21.0    18:35:52 25    6    10394- 712M- 17M    266M    587    1    sleeping *0[1]    0.16991
593    coreaudioid 15.4    93:06:46 15    3    8753    40M    0B      24M     593    1    sleeping *0[1]    0.00000
85893  promecefplug 14.6    38:44:61 22    1    253    46M    0B      20M     32270 32270 sleeping *0[1]    0.00000
96893  Google Chrom 11.8    24:14:27 19    1    272    28M    0B      18M     96762 96762 sleeping *0[13935] 0.05544
0      kernel_task  9.6     13:37:12 622/10 0    0      15M+   0B      0B      0      0    running  0[0]    0.00000
72731  top          9.5     00:01:52 1/1    0    30-    10M    0B      0B      72731 70833 running  *0[1]    0.00000
74493  promecefplug 3.6     00:23:09 16    1    307    66M    0B      35M-    32270 32270 sleeping *0[1]    0.00000
554    launchservic 2.7     59:15:58 8      7    1569- 11M    0B      3568K    554    1    sleeping *50465[247] 2.69201
81723  WhatsApp     2.5     03:18:11 36    5    3058    1291M 1344K    880M    81723 1    sleeping *1866[31] 0.00000
41981  wpsbackend   2.4     35:32:59 29    1    241    50M    0B      41M     41981 32270 sleeping *0[3808] 0.00000
586    loginwindow  2.3     94:52:73 5      2    826    85M    0B      105M    586    1    sleeping *1974[3460] 0.00000
96762  Google Chrom 2.1     62:49:15 47    3    1849    382M    0B      186M    96762 1    sleeping *4[4295] 0.03865
679    com.apple.Dr 1.6     01:56:42 7/1    5    1405    32M    0B      4816K    679    1    running  0[1]    0.00000
96804  Google Chrom 1.5     24:10:07 19    1    156    53M    0B      26M     96762 96762 sleeping *0[3]    0.00000
576    bluetoothd   1.1     64:44:55 13    5    558    17M    240K    6672K    576    1    sleeping *0[1]    0.00000
2695  Codex (Servi 1.0     46:52:51 23    3    232    212M    0B      33M     2667 2667 sleeping *1[503] 0.00000
81796  Notes        0.9     11:31:55 9      5    625    77M    0B      56M     81796 1    sleeping *0[9688] 0.59969
29875  Discord Help 0.6     31:46:22 45    1    795    364M+ 0B      122M-    29847 29847 sleeping *3[47578] 0.16493
87577  Google Chrom 0.5     16:28:25 34    1    443    332M    0B      181M    96762 96762 sleeping 0[86]    0.00000
```

Conclusion:

WSL installation is not applicable to mac-OS because WSL is a Windows-specific technology. The required Unix/Linux-based OS practical commands can be performed directly through the mac-OS Terminal.

Therefore, the Terminal screenshots provided above serve as evidence of the practical work performed on mac-OS.

SIMPLE COMMANDS:

TOPIC 2 : Directory Management

1. ls

```
Last login: Wed Aug 12 17:52:31 on ttys007
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_api      Documents          Music              Public
abc.c              bookflow_fastapi  Downloads          mysql              PyCharmMiscProject
abc.c#y            CREATE            dsa 3 odd sem     OSSP              PycharmProjects
Applications       data              Library            package-lock.json  show
blockhack hackathon Desktop            Movies             Pictures
amrutha@Amruthas-MacBook-Air ~ %
```

2. ls-l

```
amrutha@Amruthas-MacBook-Air ~ % ls -l
total 144
-rwxr-xr-x  1 amrutha  staff  33424 Jul 31 13:54 abc
-rw-r--r--  1 amrutha  staff    80 Jul 31 13:51 abc.c
-rw-r--r--  1 amrutha  staff    81 Jul 29 12:39 abc.c#y
drwx-----@ 5 amrutha  staff   160 Jun 19 10:23 Applications
drwxr-xr-x  2 amrutha  staff    64 Aug  7 09:39 blockhack hackathon
drwxr-xr-x  3 amrutha  staff    96 Aug 12 15:31 bookflow_api
drwxr-xr-x  5 amrutha  staff   160 Aug 12 17:20 bookflow_fastapi
-rw-r--r--  1 amrutha  staff  17827 Jul 30 11:02 CREATE
drwxr-xr-x  3 amrutha  staff    96 Aug  3 14:53 data
drwx-----@ 138 amrutha  staff   4416 Aug 12 17:59 Desktop
drwx-----@  5 amrutha  staff   160 Aug 11 13:52 Documents
drwx-----@ 337 amrutha  staff  10784 Aug 12 18:09 Downloads
drwxr-xr-x  3 amrutha  staff    96 Jul 23 12:20 dsa 3 odd sem
drwx-----@ 91 amrutha  staff   2912 Jul 25 08:50 Library
drwx-----  5 amrutha  staff   160 Apr 22 21:03 Movies
drwx-----+ 4 amrutha  staff   128 Mar 20 10:47 Music
-rw-r--r--  1 amrutha  staff    51 Jul 30 11:02 mysql
drwxr-xr-x 17 amrutha  staff   544 Aug  3 12:41 OSSP
-rw-r--r--@  1 amrutha  staff    86 May 11 10:41 package-lock.json
drwx-----+ 6 amrutha  staff   192 Jun 13 20:31 Pictures
drwxr-xr-x+ 4 amrutha  staff   128 Feb 21 12:34 Public
drwxr-xr-x@ 4 amrutha  staff   128 Apr 16 14:18 PyCharmMiscProject
drwxr-xr-x@ 3 amrutha  staff    96 Apr 16 14:18 PycharmProjects
-rw-r--r--  1 amrutha  staff     0 Jul 30 11:03 show
amrutha@Amruthas-MacBook-Air ~ %
```

3. ls -a

```
amrutha@Amruthas-MacBook-Air ~ % ls -a
.          .DS_Store          .npmrc          abc.c          Movies
..         .gemini           .oracle_jre_usage abc.c#y        Music
.antigravity .gitconfig       .psql_history   Applications   mysql
.antigravity-ide .homebrew       .redhat         blockhack hackathon OSSP
.anydesk     .idlerc         .ssh            bookflow_api   package-lock.json
.arduinoIDE  .ipython        .Trash          bookflow_fastapi Pictures
.bash_history .junie          .vscode         CREATE         Public
.cache       .Library        .vscode-shared  data          PyCharmMiscProject
.CFUserTextEncoding .lima          .zprofile       Desktop       PyCharmProjects
.claude      .local          .zprofile.pysave Documents      show
.codex       .mongodb        .zsh_history    Downloads
.config     .mysql_history  .zsh_sessions  dsa 3 odd sem
.copilot    .npm            abc             Library
```

4. cd

```
amrutha@Amruthas-MacBook-Air ~ % pwd
/Users/amrutha
amrutha@Amruthas-MacBook-Air ~ % cd ~/Documents
amrutha@Amruthas-MacBook-Air Documents % pwd
/Users/amrutha/Documents
```

5. cd ..

```
amrutha@Amruthas-MacBook-Air Documents % cd ..
amrutha@Amruthas-MacBook-Air ~ % pwd
/Users/amrutha
amrutha@Amruthas-MacBook-Air ~ %
```

6. mkdir

```
amrutha@Amruthas-MacBook-Air ~ % mkdir testfolder
amrutha@Amruthas-MacBook-Air ~ % ls
abc          bookflow_api      Documents      Music          Public
abc.c        bookflow_fastapi  Downloads     mysql          PyCharmMiscProject
abc.c#y      CREATE           dsa 3 odd sem OSSP           PyCharmProjects
Applications data             Library       package-lock.json show
blockhack hackathon Desktop         Movies        Pictures       testfolder
```

7. rmdir

```
amrutha@Amruthas-MacBook-Air ~ % rmdir testfolder
amrutha@Amruthas-MacBook-Air ~ % ls
abc          bookflow_api      Documents      Music          Public
abc.c        bookflow_fastapi  Downloads     mysql          PyCharmMiscProject
abc.c#y      CREATE           dsa 3 odd sem OSSP           PyCharmProjects
Applications data             Library       package-lock.json show
blockhack hackathon Desktop         Movies        Pictures
```

8. find

```
amrutha@Amruthas-MacBook-Air ~ % mkdir -p ~/os_test
amrutha@Amruthas-MacBook-Air ~ % touch ~/os_test/sample.txt
amrutha@Amruthas-MacBook-Air ~ % find ~ -name "sample.txt"
find: /Users/amrutha/Music/Music: Operation not permitted
find: /Users/amrutha/Pictures/Photos Library.photoslibrary: Operation not permitted
/Users/amrutha/Desktop/TextHack/sample.txt
find: /Users/amrutha/Library/Application Support/CallHistoryTransactions: Operation not permitted
find: /Users/amrutha/Library/Application Support/CloudDocs: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.sharedfilelist: Operation not permitted
find: /Users/amrutha/Library/Application Support/Knowledge: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.TCC: Operation not permitted
find: /Users/amrutha/Library/Application Support/FileProvider: Operation not permitted
find: /Users/amrutha/Library/Application Support/AddressBook: Operation not permitted
find: /Users/amrutha/Library/Application Support/FaceTime: Operation not permitted
find: /Users/amrutha/Library/Application Support/DifferentialPrivacy: Operation not permitted
find: /Users/amrutha/Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
find: /Users/amrutha/Library/Application Support/CallHistoryDB: Operation not permitted
find: /Users/amrutha/Library/Assistant/SiriVocabulary: Operation not permitted
find: /Users/amrutha/Library/Daemon Containers: Operation not permitted
find: /Users/amrutha/Library/Autosave Information: Operation not permitted
```

9. locate

```
amrutha@Amruthas-MacBook-Air ~ % locate sample.txt

WARNING: The locate database (/var/db/locate.database) does not exist.
To create the database, run the following command:

    sudo launchctl load -w /System/Library/LaunchDaemons/com.apple.locate.plist

Please be aware that the database can take some time to generate; once
the database has been created, this message will no longer appear.

amrutha@Amruthas-MacBook-Air ~ % mdfind "sample.txt"
2026-08-12 18:52:35.453 mdfind[50593:9947903] [UserQueryParser] Loading keywords and predicates for locale "en_US"
2026-08-12 18:52:35.453 mdfind[50593:9947903] [UserQueryParser] Loading keywords and predicates for locale "en"
/Users/amrutha/os_test/sample.txt
```

TOPIC 3 : File Management

1. touch

```
amrutha@Amruthas-MacBook-Air ~ % touch
usage: touch [-A [-l[hh]mm]SS] [-achm] [-r file] [-t [[CC]YY]MMDDhhmm[.SS]]
        [-d YYYY-MM-DDThh:mm:SS[.frac][tz]] file ...
amrutha@Amruthas-MacBook-Air ~ % touch test.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    dsa 3 odd sem      OSSP                show
abc.c              CREATE              Library            package-lock.json   test.txt
abc.c#y            data                Movies             Pictures
Applications       Desktop             Music              Public
blockhack hackathon Documents           mysql               PyCharmMiscProject
bookflow_api        Downloads           os_test            PycharmProjects
amrutha@Amruthas-MacBook-Air ~ %
```

2. cat

```
amrutha@Amruthas-MacBook-Air ~ % echo "Hello from my Mac" > test.txt
amrutha@Amruthas-MacBook-Air ~ % cat test.txt
Hello from my Mac
amrutha@Amruthas-MacBook-Air ~ % cp test.txt copy.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    Downloads           os_test              PycharmProjects
abc.c              copy.txt            dsa 3 odd sem      OSSP                show
abc.c#y            CREATE              Library            package-lock.json   test.txt
Applications       data                Movies             Pictures
Applications       Desktop             Music              Public
blockhack hackathon Documents           mysql               PyCharmMiscProject
bookflow_api        Downloads
amrutha@Amruthas-MacBook-Air ~ %
```

3. cp

```
amrutha@Amruthas-MacBook-Air ~ % cp test.txt copy.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    Downloads          os_test            PycharmProjects
abc.c              copy.txt            dsa 3 odd sem     OSSP              show
abc.c#y           CREATE             Library            package-lock.json  test.txt
Applications       data               Movies
blockhack hackathon Desktop            Music
bookflow_api      Documents          mysql
amrutha@Amruthas-MacBook-Air ~ %
```

4. mv

```
amrutha@Amruthas-MacBook-Air ~ % mv copy.txt renamed.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    dsa 3 odd sem     OSSP              renamed.txt
abc.c              CREATE             Library            package-lock.json  show
abc.c#y           data               Movies             Pictures           test.txt
Applications       Desktop            Music
blockhack hackathon Documents          mysql              PyCharmMiscProject
bookflow_api      Downloads          os_test            PycharmProjects
amrutha@Amruthas-MacBook-Air ~ %
```

5. rm

```
amrutha@Amruthas-MacBook-Air ~ % rm renamed.txt
amrutha@Amruthas-MacBook-Air ~ % ls
abc                bookflow_fastapi    dsa 3 odd sem     OSSP              show
abc.c              CREATE             Library            package-lock.json  test.txt
abc.c#y           data               Movies
Applications       Desktop            Music
blockhack hackathon Documents          mysql
bookflow_api      Downloads          os_test            PyCharmMiscProject
amrutha@Amruthas-MacBook-Air ~ %
```

6. head

```
amrutha@Amruthas-MacBook-Air ~ % head test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
amrutha@Amruthas-MacBook-Air ~ % head -5 test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
amrutha@Amruthas-MacBook-Air ~ %
```

7.tail

```
[amrutha@Amruthas-MacBook-Air ~ % tail test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
[amrutha@Amruthas-MacBook-Air ~ % tail -5 test.txt
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
amrutha@Amruthas-MacBook-Air ~ % █
```

8.more

```
amrutha@Amruthas-MacBook-Air ~ % more test.txt
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
```

9.less

```
Line 1: Apple
Line 2: Banana
Line 3: Mango
Line 4: Orange
Line 5: Grapes
Line 6: Watermelon
Line 7: Pineapple
Line 8: Papaya
Line 9: Strawberry
Line 10: Guava
```

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

(END)

10. wc

```
Last login: Wed Aug 12 19:23:32 on ttys007
amrutha@Amruthas-MacBook-Air ~ % wc test.txt
  10      30     159 test.txt
amrutha@Amruthas-MacBook-Air ~ % wc -l test.txt
  10 test.txt
amrutha@Amruthas-MacBook-Air ~ % wc -w test.txt
   30 test.txt
amrutha@Amruthas-MacBook-Air ~ % wc -c test.txt
  159 test.txt
amrutha@Amruthas-MacBook-Air ~ %
```

11. diff

```
[amrutha@Amruthas-MacBook-Air ~ % cp test.txt test2.txt
[amrutha@Amruthas-MacBook-Air ~ % echo "Line 11: Kiwi" >> test2.txt
[amrutha@Amruthas-MacBook-Air ~ % diff test.txt test2.txt
10a11
> Line 11: Kiwi
amrutha@Amruthas-MacBook-Air ~ %
```

12. sort

```
amrutha@Amruthas-MacBook-Air ~ % echo "Mango
Apple
Banana
Apple
Orange
[Banana" > names.txt
[amrutha@Amruthas-MacBook-Air ~ % cat names.txt
Mango
Apple
Banana
Apple
Orange
Banana
[amrutha@Amruthas-MacBook-Air ~ % sort names.txt
Apple
Apple
Banana
Banana
Mango
Orange
amrutha@Amruthas-MacBook-Air ~ %
```

13. uniq

```
[amrutha@Amruthas-MacBook-Air ~ % sort names.txt | uniq
Apple
Banana
Mango
Orange
```

14. grep

```
amrutha@Amruthas-MacBook-Air ~ % cat names.txt
Mango
Apple
Banana
Apple
Orange
Banana
amrutha@Amruthas-MacBook-Air ~ % grep "Apple" names.txt
Apple
Apple
amrutha@Amruthas-MacBook-Air ~ %
```

TOPIC 4 : File Editing

1. nano

```
amrutha@Amruthas-MacBook-Air ~ % nano editor_test.txt
amrutha@Amruthas-MacBook-Air ~ % cat editor_test.txt
Hello, this file was created using Nano.
amrutha@Amruthas-MacBook-Air ~ %
```

2. vi

```
Last login: Wed Aug 12 19:26:46 on ttys007
amrutha@Amruthas-MacBook-Air ~ % vi editor_test.txt
amrutha@Amruthas-MacBook-Air ~ % cat editor_test.txt
Hello, this file was created using Nano.
amrutha@Amruthas-MacBook-Air ~ %
```

TOPIC 5 : C Program Compilation and Execution

1. gcc

```
amrutha@Amruthas-MacBook-Air ~ % nano hello.c
amrutha@Amruthas-MacBook-Air ~ % gcc hello.c -o hello
amrutha@Amruthas-MacBook-Air ~ %
```

2. ./program

```
amrutha@Amruthas-MacBook-Air ~ % ./hello
Hello from C program!
amrutha@Amruthas-MacBook-Air ~ %
```

3. which

```
amrutha@Amruthas-MacBook-Air ~ % which gcc
/usr/bin/gcc
```

4. whereis

```
amrutha@Amruthas-MacBook-Air ~ % whereis
usage: whereis [-abmqu] [-BM dir ... -f] program ...
amrutha@Amruthas-MacBook-Air ~ % whereis gcc
gcc: /usr/bin/gcc
amrutha@Amruthas-MacBook-Air ~ % which
amrutha@Amruthas-MacBook-Air ~ %
```

5. man

```
CLANG(1)                                Clang                                CLANG(1)

NAME
    clang - the Clang C, C++, and Objective-C compiler

SYNOPSIS
    clang [options] filename ...

DESCRIPTION
    clang is a C, C++, and Objective-C compiler which encompasses preprocessing, parsing, optimization, code generation, assembly, and linking. Depending on which high-level mode setting is passed, Clang will stop before doing a full link. While Clang is highly integrated, it is important to understand the stages of compilation, to understand how to invoke it. These stages are:

    Driver The clang executable is actually a small driver which controls the overall execution of other tools such as the compiler, assembler and linker. Typically you do not need to interact with the driver, but you transparently use it to run the other tools.

    Preprocessing
        This stage handles tokenization of the input source file, macro expansion, #include expansion and handling of other preprocessor directives. The output of this stage is typically called a ".i" (for C), ".ii" (for C++), ".mi" (for Objective-C), or ".mii" (for Objective-C++) file.

    Parsing and Semantic Analysis
        This stage parses the input file, translating preprocessor tokens into a parse tree. Once in the form of a parse tree, it applies semantic analysis to compute types for expressions as well and determine whether the code is well formed. This stage is responsible for generating most of the compiler warnings as well as parse errors. The output of this stage is an "Abstract Syntax Tree" (AST).
```

TOPIC 6 : Process Management

1. ps

```
amrutha@Amruthas-MacBook-Air ~ % ps
  PID TTY          TIME CMD
 60296 ttys007    0:00.02 -zsh
amrutha@Amruthas-MacBook-Air ~ %
```

2. ps -ef

```
amrutha@Amruthas-MacBook-Air ~ % ps -ef
UID      PID  PPID  C  STIME  TTY          TIME CMD
0         1     0  0  16Jul26 ??        77:09.02 /sbin/launchd
0        511     1  0  16Jul26 ??        35:04.19 /usr/libexec/logd
0        513     1  0  16Jul26 ??        1:52.77 /usr/libexec/UserEventAgent (System)
0        515     1  0  16Jul26 ??        23:44.05 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/FSEvents.framework/Versions/A/Support/fsevents
0        516     1  0  16Jul26 ??        1:30.71 /System/Library/PrivateFrameworks/MediaRemote.framework/Support/mediare
moted
0        519     1  0  16Jul26 ??        0:45.03 /usr/sbin/systemstats --daemon
0        523     1  0  16Jul26 ??        7:08.94 /usr/libexec/configd
0        525     1  0  16Jul26 ??        5:01.31 /System/Library/CoreServices/powerd.bundle/powerd
0        526     1  0  16Jul26 ??        0:00.03 /usr/libexec/IOMFB_bics_daemon
0        530     1  0  16Jul26 ??        0:00.39 /usr/libexec/remoted
306      534     1  0  16Jul26 ??        0:00.03 /System/Library/PrivateFrameworks/CoreSpeech.framework/corespeechd_syst
em
0        536     1  0  16Jul26 ??        0:13.09 /usr/libexec/watchdogd
0        540     1  0  16Jul26 ??        28:57.18 /System/Library/Frameworks/CoreServices.framework/Versions/A/Frameworks
/Metadata.framework/Versions/A/Support/mds
0        542     1  0  16Jul26 ??        0:01.33 /usr/libexec/kernelmanagerd
0        543     1  0  16Jul26 ??        4:39.14 /usr/libexec/diskarbitrationd
0        546     1  0  16Jul26 ??        4:50.82 /usr/libexec/coreduetd
0        547     1  0  16Jul26 ??        0:52.38 /usr/sbin/syslogd
0        549     1  0  16Jul26 ??        0:53.66 /usr/libexec/thermalmonitord
0        550     1  0  16Jul26 ??        4:25.35 /System/Library/PrivateFrameworks/CoreDuetContext.framework/Resources/c
ontextstored
0        551     1  0  16Jul26 ??        4:00.00 /usr/libexec/xprotectd
0        552     1  0  16Jul26 ??        27:35.38 /usr/libexec/opendirectoryd
0        553     1  0  16Jul26 ??        2:33.68 /System/Library/PrivateFrameworks/ApplePushService.framework/apsd
0        554     1  0  16Jul26 ??        58:23.08 /System/Library/CoreServices/launchservicesd
```

3. top

```
Processes: 819 total, 3 running, 816 sleeping, 4404 threads
Load Avg: 2.66, 2.97, 3.12 CPU usage: 16.62% user, 6.14% sys, 77.23% idle
SharedLibs: 642M resident, 127M data, 93M linkedit. MemRegions: 0 total, 0B resident, 0B private, 1646M shared.
PhysMem: 15G used (2476M wired, 4767M compressor), 107M unused.
VM: 370T vsize, 6144M framework vsize, 4471(0) swapins, 75960(0) swapouts.
Networks: packets: 211970048/93G in, 21412992/15G out. Disks: 35201957/790G read, 20322694/364G written.
19:59:21

PID    COMMAND      %CPU    TIME    #TH    #WQ    #PORTS  MEM     PURG    CMPRS    PGRP    PPID    STATE    BOOSTS    %CPU_ME
587    WindowServer 51.4    18:11:48 24     5      10396+  710M-  16M-    252M    587     1      sleeping *0[1]    1.43899
48745  Google Chrom 35.8    29:18:03 42/1    1      561-    599M+  0B      250M-   96762  96762  running *0[7]    0.00000
32277  promceefplug 24.4    10:45:01 14      5      252     228M   0B      49M     32270  32270  sleeping *1[5]    0.08337
32270  wpsoffice    20.6    09:19:09 82      1      1741+   1327M  112K    896M    32270  1      sleeping *1[16416] 11.7587
85893  promceefplug 14.0    31:46:04 22      1      253     47M+   0B      21M     32270  32270  sleeping *0[1]    0.00000
593    coreaudioid 13.5    85:09:04 14      2      8670    40M    0B      25M     593     1      sleeping *0[1]    0.15385
0      kernel_task  11.0    13:28:37 622/11  0      0       15M    0B      0B      0       0      running  0[0]    0.00000
96893  Google Chrom 9.8     19:00:02 19      1      271     28M    0B      18M     96762  96762  sleeping *0[13701] 0.03369
61795  top          9.2     00:01:95 1/1     0      30      10M    0B      0B      61795  60296  running *0[1]    0.00000
96803  Google Chrom 7.1     01:54:49 25      3      502     608M+  32M-    136M    96762  96762  sleeping *1[31603] 0.76359
96762  Google Chrom 5.3     59:56:32 46      2      1851    381M+  0B      184M-   96762  1      sleeping *4[4295] 0.04954
87577  Google Chrom 4.0     15:36:65 34      1      441     340M-  0B      165M-   96762  96762  sleeping 0[81]    0.00000
41981  wpsbackend   2.1     34:25:78 29      1      243     48M    0B      37M     41981  32270  sleeping *0[3468] 0.00000
81723  WhatsApp     2.1     03:01:39 36      5      3065-   1269M- 1440K    792M-   81723  1      sleeping *1846[31] 0.00000
470    con.apple.Ds 1.0     01:55:00 7        5      1405    32M    0B      1734K   470     1      sleeping 0[1]    0.00000
```

4. kill

```
amrutha@Amruthas-MacBook-Air ~ % sleep 1000 &
[2] 62597
[3] 62757
zsh: no matches found: [2]
amrutha@Amruthas-MacBook-Air ~ % ps -p 62597
  PID TTY          TIME CMD
 62597 ttys007      0:00.00 sleep 1000
amrutha@Amruthas-MacBook-Air ~ % kill 62597
amrutha@Amruthas-MacBook-Air ~ %
[2] - terminated  sleep 1000
amrutha@Amruthas-MacBook-Air ~ % ps -p 62597
  PID TTY          TIME CMD
```

5. kill -9

```
amrutha@Amruthas-MacBook-Air ~ % sleep 1000 &
[2] 63098
amrutha@Amruthas-MacBook-Air ~ % ps -p 63025
  PID TTY          TIME CMD
amrutha@Amruthas-MacBook-Air ~ %
amrutha@Amruthas-MacBook-Air ~ % kill -9 63025
kill: kill 63025 failed: no such process
amrutha@Amruthas-MacBook-Air ~ % ps -p 63025
  PID TTY          TIME CMD
amrutha@Amruthas-MacBook-Air ~ %
```

Topic 7: Job Control

1. jobs

```
amrutha@Amruthas-MacBook-Air ~ % jobs
[1]  running    sleep 1000
[2]  running    sleep 1000
[3]  - running    sleep 1000
[4]  + running    sleep 1000
amrutha@Amruthas-MacBook-Air ~ %
```

2. bg

```
[amrutha@Amruthas-MacBook-Air ~ % sleep 1000
^Z
zsh: suspended  sleep 1000
[amrutha@Amruthas-MacBook-Air ~ % bg
[5] - continued  sleep 1000
[amrutha@Amruthas-MacBook-Air ~ % jobs
[1]  running    sleep 1000
[2]  running    sleep 1000
[3]  running    sleep 1000
[4] + running    sleep 1000
[5] - running    sleep 1000
amrutha@Amruthas-MacBook-Air ~ %
```

3. fg

```
[amrutha@Amruthas-MacBook-Air ~ % fg
[4] - running    sleep 1000
^C
amrutha@Amruthas-MacBook-Air ~ %
```

4. &

```
amrutha@Amruthas-MacBook-Air ~ % sleep 1000 &
[1] 62430
```

Topic 8: File Permissions and Links

1. chmod

```
[amrutha@Amruthas-MacBook-Air ~ % touch permission.txt
[amrutha@Amruthas-MacBook-Air ~ % ls -l permission.txt
-rw-r--r--  1 amrutha  staff   0 Aug 12 20:15 permission.txt
[amrutha@Amruthas-MacBook-Air ~ % chmod 644 permission.txt
[amrutha@Amruthas-MacBook-Air ~ % ls -l permission.txt
-rw-r--r--  1 amrutha  staff   0 Aug 12 20:15 permission.txt
amrutha@Amruthas-MacBook-Air ~ %
```

2. chmod 755

```
amrutha@Amruthas-MacBook-Air ~ % chmod 755 permission.txt
amrutha@Amruthas-MacBook-Air ~ % ls -l permission.txt
-rwxr-xr-x 1 amrutha staff 0 Aug 12 20:15 permission.txt
amrutha@Amruthas-MacBook-Air ~ %
```

3. ln

```
amrutha@Amruthas-MacBook-Air ~ % echo "Hello from original file" > original.txt
amrutha@Amruthas-MacBook-Air ~ %
[1] done sleep 1000
amrutha@Amruthas-MacBook-Air ~ % ln original.txt hardlink.txt
amrutha@Amruthas-MacBook-Air ~ % ls -l original.txt hardlink.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 hardlink.txt
-rw-r--r-- 2 amrutha staff 25 Aug 12 20:19 original.txt
amrutha@Amruthas-MacBook-Air ~ % cat hardlink.txt
Hello from original file
amrutha@Amruthas-MacBook-Air ~ %
```

4. ln -s

```
amrutha@Amruthas-MacBook-Air ~ % ln -s original.txt symlink.txt
amrutha@Amruthas-MacBook-Air ~ % ls -l symlink.txt
lrwxr-xr-x 1 amrutha staff 12 Aug 12 20:20 symlink.txt -> original.txt
amrutha@Amruthas-MacBook-Air ~ % cat symlink.txt
Hello from original file
amrutha@Amruthas-MacBook-Air ~ %
```

Topic 9: Disk and Memory Information

1. df -h

```
amrutha@Amruthas-MacBook-Air ~ % df -h
Filesystem      Size  Used Avail Capacity iused ifree %iused  Mounted on
/dev/disk3s1s1  460Gi  12Gi  318Gi    4%   459k  3.3G    0%      /
devfs           213Ki  213Ki   0Bi  100%    736    0  100%    /dev
/dev/disk3s6    460Gi  2.0Gi  318Gi    1%      2  3.3G    0%    /System/Volumes/VM
/dev/disk3s2    460Gi  8.4Gi  318Gi    3%   1.4k  3.3G    0%    /System/Volumes/Preboot
/dev/disk3s4    460Gi  2.4Mi  318Gi    1%     65  3.3G    0%    /System/Volumes/Update
/dev/disk1s2    500Mi  6.0Mi  482Mi    2%      1  4.9M    0%    /System/Volumes/xarts
/dev/disk1s1    500Mi  6.0Mi  482Mi    2%     33  4.9M    0%    /System/Volumes/iSCPreboot
/dev/disk1s3    500Mi  1.4Mi  482Mi    1%     94  4.9M    0%    /System/Volumes/Hardware
/dev/disk3s5    460Gi  119Gi  318Gi   28%   1.3M  3.3G    0%    /System/Volumes/Data
map auto_home   0Bi    0Bi   0Bi  100%      0    0    -    /System/Volumes/Data/home
/dev/disk5s1    677Mi  525Mi  148Mi   78%   7.9k  1.5M    1%    /Volumes/Arduino IDE 2.3.10-arm64
/dev/disk6s1    1.3Gi  1.3Gi  2.4Mi  100%    11k  4.3G    0%    /Volumes/ChatGPT Installer
amrutha@Amruthas-MacBook-Air ~ %
```

2. du -sh

```
amrutha@Amruthas-MacBook-Air ~ % du -sh .
du: ./Music/Music: Operation not permitted
du: ./Pictures/Photos Library.photoslibrary: Operation not permitted
du: ./Library/Application Support/CallHistoryTransactions: Operation not permitted
du: ./Library/Application Support/CloudDocs: Operation not permitted
du: ./Library/Application Support/com.apple.sharedfilelist: Operation not permitted
du: ./Library/Application Support/Knowledge: Operation not permitted
du: ./Library/Application Support/com.apple.TCC: Operation not permitted
du: ./Library/Application Support/FileProvider: Operation not permitted
du: ./Library/Application Support/AddressBook: Operation not permitted
du: ./Library/Application Support/FaceTime: Operation not permitted
du: ./Library/Application Support/DifferentialPrivacy: Operation not permitted
du: ./Library/Application Support/com.apple.avfoundation/Frecents: Operation not permitted
du: ./Library/Application Support/CallHistoryDB: Operation not permitted
du: ./Library/Assistant/SiriVocabulary: Operation not permitted
du: ./Library/Daemon Containers: Operation not permitted
du: ./Library/Autosave Information: Operation not permitted
du: ./Library/IdentityServices: Operation not permitted
du: ./Library/Messages: Operation not permitted
du: ./Library/HomeKit: Operation not permitted
du: ./Library/Sharing: Operation not permitted
du: ./Library/com.apple.aiml.instrumentation: Operation not permitted
du: ./Library/Mail: Operation not permitted
du: ./Library/Trial: Operation not permitted
du: ./Library/AppleMediaServices: Operation not permitted
du: ./Library/DuetExpertCenter: Operation not permitted
du: ./Library/Accounts: Operation not permitted
du: ./Library/Safari: Operation not permitted
du: ./Library/Bioime: Operation not permitted
du: ./Library/IntelligencePlatform: Operation not permitted
```

3. free -h

```
amrutha@Amruthas-MacBook-Air ~ % top -l 1 | grep PhysMem
PhysMem: 15G used (2709M wired, 5195M compressor), 177M unused.
amrutha@Amruthas-MacBook-Air ~ %
```

Topic 10: User and Environment Management

1. id

```
amrutha@Amruthas-MacBook-Air ~ % id
uid=501(amrutha) gid=20(staff) groups=20(staff),12(everyone),61(localaccounts),79(_appserverusr),80(admin),81(_appserver
adm),33(_appstore),98(_lpadmin),100(_lpoperator),204(_developer),250(_analyticsusers),395(com.apple.access_ftp),398(com.
apple.access_screensharing),399(com.apple.access_ssh),400(com.apple.access_remote_ae),701(com.apple.sharepoint.group.1)
amrutha@Amruthas-MacBook-Air ~ %
```

2. env

```

amrutha@Amruthas-MacBook-Air ~ % env
TMPDIR=/var/folders/h1/gcvz4hvxvnnv15_mqpw_qs0e0000gn/T/
CFBundleIdentifier=com.apple.Terminal
XPC_FLAGS=0x0
TERM=xterm-256color
SSH_AUTH_SOCK=/var/run/com.apple.launchd.WSKLs4dcfr/Listeners
XPC_SERVICE_NAME=0
TERM_PROGRAM=Apple_Terminal
COLORTERM=truecolor
OSLogRateLimit=64
TERM_PROGRAM_VERSION=470.2
TERM_SESSION_ID=862A7243-698C-4EB3-99B4-AF7DMC18D51E
SHELL=/bin/zsh
HOME=/Users/amrutha
LOGNAME=amrutha
USER=amrutha
PATH=/opt/homebrew/opt/postgresql@17/bin:/opt/homebrew/bin:/opt/homebrew/sbin:/Library/Frameworks/Python.framework/Versions/3.14/bin:/usr/local/bin:/System/Cryptexes/App/usr/bin:/usr/bin:/bin:/usr/sbin:/sbin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/local/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/bin:/var/run/com.apple.security.cryptexd/codex.system/bootstrap/usr/appleinternal/bin:/pkg/env/global/bin
SHELLVL=1
PWD=/Users/amrutha
OLDPWD=/Users/amrutha
HOMEBREW_PREFIX=/opt/homebrew
HOMEBREW_CELLAR=/opt/homebrew/Cellar
HOMEBREW_REPOSITORY=/opt/homebrew
PPATH=/opt/homebrew/share/zsh/site-functions:/usr/local/share/zsh/site-functions:/usr/share/zsh/site-functions:/usr/share/zsh/5.9/functions
INFOPATH=/opt/homebrew/share/info:
LANG=en_US.UTF-8
_=/usr/bin/env
amrutha@Amruthas-MacBook-Air ~ %

```

3. export

```

amrutha@Amruthas-MacBook-Air ~ % export MYNAME="Amrutha"
amrutha@Amruthas-MacBook-Air ~ % echo $MYNAME
Amrutha
amrutha@Amruthas-MacBook-Air ~ %

```

4. alias

```

amrutha@Amruthas-MacBook-Air ~ % alias ll='ls -l'
amrutha@Amruthas-MacBook-Air ~ % ll
total 272
-rwxr-xr-x  1 amrutha  staff  33424 Jul 31 13:54 abc
-rw-r--r--  1 amrutha  staff    80 Jul 31 13:51 abc.c
-rw-r--r--  1 amrutha  staff    81 Jul 29 12:39 abc.c#y
drwx-----@ 5 amrutha  staff   160 Jun 19 10:23 Applications
drwxr-xr-x  2 amrutha  staff    64 Aug  7 09:39 blockhack hackathon
drwxr-xr-x  3 amrutha  staff    96 Aug 12 15:31 bookflow_api
drwxr-xr-x  5 amrutha  staff   160 Aug 12 17:20 bookflow_fastapi
-rw-r--r--  1 amrutha  staff  17827 Jul 30 11:02 CREATE
drwxr-xr-x  3 amrutha  staff    96 Aug  3 14:53 data
drwx-----@ 141 amrutha  staff   4512 Aug 12 19:59 Desktop
drwx-----@  5 amrutha  staff   160 Aug 11 13:52 Documents
drwx-----@ 337 amrutha  staff  10784 Aug 12 18:09 Downloads
drwxr-xr-x  3 amrutha  staff    96 Jul 23 12:20 dsa 3 odd sem
-rw-r--r--  1 amrutha  staff    41 Aug 12 19:41 editor_test.txt
-rw-r--r--  2 amrutha  staff    25 Aug 12 20:19 hardlink.txt
-rwxr-xr-x  1 amrutha  staff  33432 Aug 12 19:43 hello
-rw-r--r--  1 amrutha  staff    88 Aug 12 19:43 hello.c
drwx-----@ 91 amrutha  staff   2912 Jul 25 08:50 Library
drwx-----  5 amrutha  staff   160 Apr 22 21:03 Movies
drwx-----+  4 amrutha  staff   128 Mar 20 10:47 Music
-rw-r--r--  1 amrutha  staff    51 Jul 30 11:02 mysql
-rw-r--r--  1 amrutha  staff    39 Aug 12 19:29 names.txt
-rw-r--r--  2 amrutha  staff    25 Aug 12 20:19 original.txt
drwxr-xr-x  3 amrutha  staff    96 Aug 12 18:50 os_test
drwxr-xr-x 17 amrutha  staff   544 Aug  3 12:41 OSSP
-rw-r--r--@  1 amrutha  staff    86 May 11 10:41 package-lock.json
-rwxr-xr-x  1 amrutha  staff     0 Aug 12 20:15 permission.txt
drwx-----+  6 amrutha  staff   192 Jun 13 20:31 Pictures
drwxr-xr-x+  4 amrutha  staff   128 Feb 21 12:34 Public
drwxr-xr-x@  4 amrutha  staff   128 Apr 16 14:18 PyCharmMiscProject
drwxr-xr-x@  3 amrutha  staff    96 Apr 16 14:18 PycharmProjects
-rw-r--r--  1 amrutha  staff     0 Jul 30 11:03 show
lrwxr-xr-x  1 amrutha  staff    12 Aug 12 20:20 symlink.txt -> original.txt
-rw-r--r--  1 amrutha  staff   159 Aug 12 19:07 test.txt
-rw-r--r--  1 amrutha  staff   173 Aug 12 19:28 test2.txt
amrutha@Amruthas-MacBook-Air ~ % █

```

(SESSION NUMBER - 2)

QUESTION 1:

TASK 1: Create the OSSP project structure using mkdir and touch commands

```
Last login: Wed Aug 12 20:53:13 on ttys007
[amrutha@Amruthas-MacBook-Air ~ % mkdir OSSP
mkdir: OSSP: File exists
[amrutha@Amruthas-MacBook-Air ~ % cd OSSP
[amrutha@Amruthas-MacBook-Air OSSP % pwd
/Users/amrutha/OSSP
[amrutha@Amruthas-MacBook-Air OSSP % mkdir src include obj bin docs tests scripts assets
mkdir: src: File exists
mkdir: include: File exists
mkdir: obj: File exists
mkdir: bin: File exists
mkdir: docs: File exists
mkdir: tests: File exists
mkdir: scripts: File exists
mkdir: assets: File exists
```

Figure 1: OSSP project directory created in mac-OS Terminal

```
[amrutha@Amruthas-MacBook-Air OSSP % find . -maxdepth 2 -not -path './.git*' -print
.
./obj
./hello.c
./LICENSE
./bin
./bin/my_shell
./Makefile
./include
./include/shell.h
./include/builtin.h
./include/parser.h
./include/executor.h
./tests
./tests/test_executor.c
./tests/test_parser.c
./docs
./docs/design.md
./docs/report.pdf
./README.md
./scripts
./scripts/run.sh
./hello
./assets
./assets/architecture.png
./src
./src/executor.c
./src/main.c.save
./src/shell.c
./src/main.c
./src/builtin.c
./src/parser.c
```

```
[amrutha@Amruthas-MacBook-Air OSSP % tree
```

```
.
├── assets
│   └── architecture.png
├── bin
│   └── my_shell
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── LICENSE
├── Makefile
├── obj
├── README.md
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c
```

```
9 directories, 19 files
```

```
amrutha@Amruthas-MacBook-Air OSSP %
```

Figure 2 & 3: OSSP project structure with source, header, documentation, test, script and asset directories

QUESTION 2:

TASK 1: Using a manual command, understand the fork() and exec() system calls. SKILL_Week_2

```
FORK(2)                                System Calls Manual                                FORK(2)

NAME
    fork - create a new process

SYNOPSIS
    #include <unistd.h>

    pid_t
    fork(void);

DESCRIPTION
    fork() causes creation of a new process. The new process (child process) is an exact copy of the calling process (parent process) except for the following:

    • The child process has a unique process ID.

    • The child process has a different parent process ID (i.e., the process ID of the parent process).

    • The child process has its own copy of the parent's descriptors. These descriptors reference the same underlying objects, so that, for instance, file pointers in file objects are shared between the child and the parent, so that an lseek(2) on a descriptor in the child process can affect a subsequent read or write by the parent. This descriptor copying is also used by the shell to establish standard input and output for newly created processes as well as to set up pipes.

    • The child processes resource utilizations are set to 0; see setrlimit(2).

RETURN VALUES
    Upon successful completion, fork() returns a value of 0 to the child process and returns the process ID of the child process to the parent process. Otherwise, a value of -1 is returned to the parent process, no child process is created, and the global variable errno is set to indicate the error.

ERRORS
    fork() will fail and no child process will be created if:

    [EAGAIN]      The system-imposed limit on the total number of processes under execution would be exceeded. This limit is configuration-dependent.

    [EAGAIN]      The system-imposed limit MAXUPRC ((sys/param.h)) on the total number of processes under execution by a single user would be exceeded.

    [ENOMEM]      There is insufficient swap space for the new process.

LEGACY SYNOPSIS
    #include <sys/types.h>
    #include <unistd.h>

    The include file <sys/types.h> is necessary.

SEE ALSO
    execve(2), sigaction(2), wait(2), compat(5)

HISTORY
    A fork() function call appeared in Version 6 AT&T UNIX.

CAVEATS
    There are limits to what you can do in the child process. To be totally safe you should restrict yourself to only executing async-signal safe operations until such time as one of the exec functions is called. All APIs, including global data symbols, in any framework or library should be assumed to be unsafe after a fork() unless
:||
```

```
EXEC(3)                                Library Functions Manual                                EXEC(3)

NAME
    execl, execlp, execlp, execv, execvp, execvp - execute a file

LIBRARY
    Standard C library (libc, -lc)

SYNOPSIS
    #include <unistd.h>

    extern char **environ;

    int
    execl(const char *path, const char *arg0, ..., /* (char *)0, */);

    int
    execlp(const char *path, const char *arg0, ..., /* (char *)0 char *const envp[] */);

    int
    execlp(const char *file, const char *arg0, ..., /* (char *)0, */);

    int
    execvp(const char *path, char *const argv[]);

    int
    execvp(const char *file, char *const argv[]);

    int
    execvp(const char *file, const char **search_path, char *const argv[]);

DESCRIPTION
    The exec family of functions replaces the current process image with a new process image. The functions described in this manual page are front-ends for the function execve(2). (See the manual page for execve(2) for detailed information about the replacement of the current process.)

    The initial argument for these functions is the pathname of a file which is to be executed.

    The const char *arg0 and subsequent ellipses in the execl(), execlp(), and execlp() functions can be thought of as arg0, arg1, ..., argn. Together they describe a list of one or more pointers to null-terminated strings that represent the argument list available to the executed program. The first argument, by convention, should point to the file name associated with the file being executed. The list of arguments must be terminated by a NULL pointer.

    The execv(), execvp(), and execvp() functions provide an array of pointers to null-terminated strings that represent the argument list available to the new program. The first argument, by convention, should point to the file name associated with the file being executed. The array of pointers must be terminated by a NULL pointer.

    The execlp() function also specifies the environment of the executed process by following the NULL pointer that terminates the list of arguments in the argument list or the pointer to the argv array with an additional argument. This additional argument is an array of pointers to null-terminated strings and must be terminated by a NULL pointer. The other functions take the environment for the new process image from the external variable environ in the current process.

    Some of these functions have special semantics.

    The functions execlp(), execvp(), and execvp() will duplicate the actions of the shell in searching for an executable file if the specified file name does not contain a slash "/" character. For execlp() and execvp(), search path is the path specified in the environment by "PATH" variable. If this variable is not specified, the default path is set according to the _PATH_DEFPATH definition in <paths.h>, which is set to "/usr/bin:/bin". For execvp(), the search path is specified as an argument to the function. In addition, certain errors are treated specially.

    If an error is ambiguous (for simplicity, we shall consider all errors except ENOEXEC as being ambiguous here, although only the critical error EACCES is really ambiguous), then these functions will act as if they stat the file to determine whether the file exists and has suitable execute permissions. If it does, they will return immediately with the global variable errno restored to the value set by execve(). Otherwise, the search will be continued. If the search completes without performing a successful execve() or terminating due to an error, these functions will return with the global variable errno set to EACCES or ENOENT according to whether at least one file with suitable execute permissions was found.
:||
```

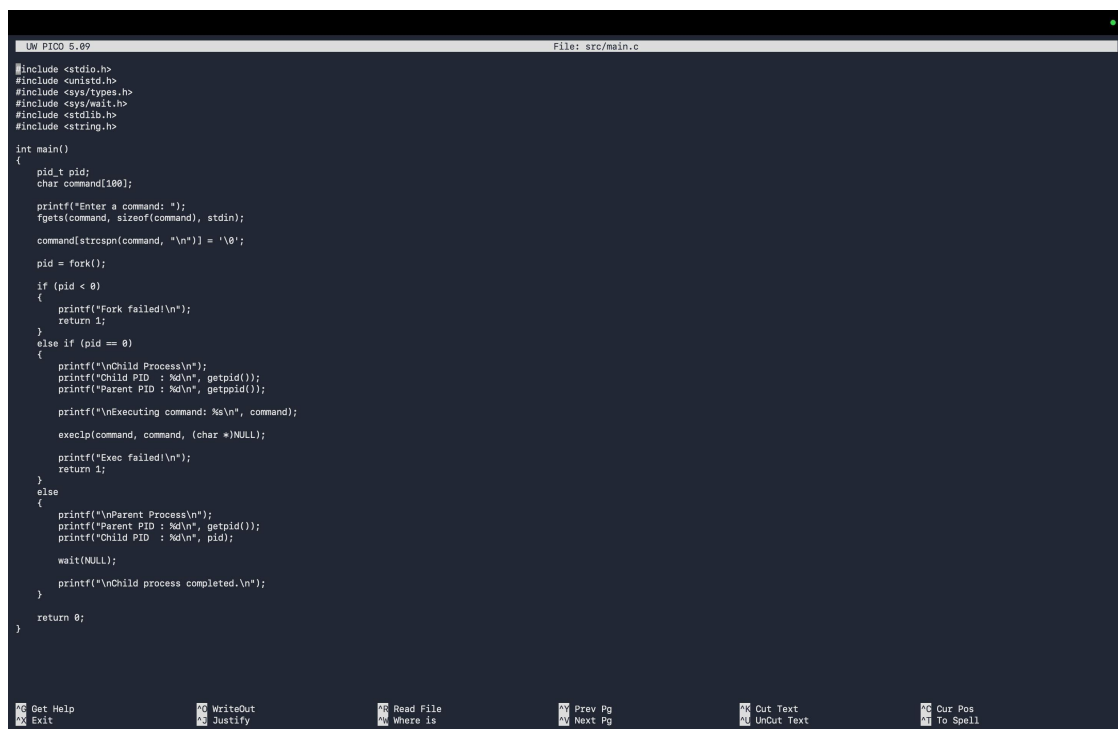
```

amrutha@Amruthas-MacBook-Air ~ % man -k fork
ResMerger(1), /usr/bin/ResMerger(1) - Merges resource forks or files into one resource file (DEPRECATED)
SplitForks(1), /usr/bin/SplitForks(1) - Divide a two-fork HFS file into AppleDouble format resource and data files. (DEPRECATED)
pg_rewind(1) - synchronize a PostgreSQL data directory with another data directory that was forked from it
OPENSSL_fork_prepare(3ssl), OPENSSL_fork_parent(3ssl), OPENSSL_fork_child(3ssl) - OpenSSL fork handlers
pg_rewind(1) - synchronize a PostgreSQL data directory with another data directory that was forked from it
perlfork(1) - Perl's fork() emulation
spfd(1) - simple forking daemon to provide SPF query services
ModPerl::PerlRunPrefork(3pm) - Run unaltered CGI scripts under mod_perl
ModPerl::RegistryPrefork(3pm) - Run unaltered CGI scripts under mod_perl
Net::Server::Daemonize(3pm) - Safe fork and daemonization utilities
Net::Server::Fork(3pm) - Net::Server personality
Net::Server::Prefork(3pm) - Net::Server personality
Net::Server::PreforkSimple(3pm) - Net::Server personality
Test2::IPC(3pm) - Turn on IPC for threading or forking support
Test2::Require::Fork(3pm) - Skip a test file unless the system supports forking
Test2::Require::RealFork(3pm) - Skip a test file unless the system supports true forking
fork(2) - create a new process
openpty(3), login_tty(3), forkpty(3) - tty utility functions
perlfork(1) - Perl's fork() emulation
pthread_atfork(3) - register handlers to be called before and after fork
spfd(1) - simple forking daemon to provide SPF query services
vfork(2) - deprecated system call to create a new process

```

Figure: Manual search for fork() system call using man -k fork

Task 2: Write a C program to create a new process and accept user commands using fork() and exec() system calls.



```

File: src/main.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter a command: ");
    fgets(command, sizeof(command), stdin);
    command[strcspn(command, "\n")] = '\0';

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("\nExecuting command: %s\n", command);

        execlp(command, command, (char *)NULL);

        printf("Exec failed!\n");
        return 1;
    }
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}

```

Figure 1: C program implementing fork() and exec()

```

[amrutha@Amruthas-MacBook-Air OSSP % gcc src/main.c -o bin/my_shell
amrutha@Amruthas-MacBook-Air OSSP %

```

Figure 2: Successful compilation of the C program

```

amrutha@Amruthas-MacBook-Air OSSP % ./bin/my_shell
Enter a command: ls

Parent Process
Parent PID : 38663
Child PID  : 38679

Child Process
Child PID  : 38679
Parent PID : 38663

Executing command: ls
assets      bin         docs        include     LICENSE     Makefile    obj         README.md  scripts    src         tests

Child process completed.
amrutha@Amruthas-MacBook-Air OSSP %

```

Figure 3: Execution of ls command

```

[amrutha@Amruthas-MacBook-Air OSSP % ./bin/my_shell
Enter a command: pwd

Parent Process
Parent PID : 39007
Child PID  : 39018

Child Process
Child PID  : 39018
Parent PID : 39007

Executing command: pwd
/Users/amrutha/OSSP

Child process completed.
amrutha@Amruthas-MacBook-Air OSSP %

```

Figure 4: Execution of pwd command

fork()

pid = fork();

Creates a new child process.

exec()

execlp(command, command, (char *)NULL);

Replaces the child process with the program corresponding to the command entered by the user.

wait()

wait(NULL);

Makes the parent process wait for the child process to finish.

getpid()

getpid()

Returns the PID of the current process.

getppid()

getppid()

Returns the PID of the parent process.

```
amrutha@Amruthas-MacBook-Air OSSP % cd ~/OSSP
amrutha@Amruthas-MacBook-Air OSSP % git status
On branch main
Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   bin/my_shell
        deleted:    hello
        deleted:    hello.c
        modified:   src/main.c
        deleted:    src/main.c.save

no changes added to commit (use "git add" and/or "git commit -a")
```

Figure 5: Git status showing changes made for Task 2 and removal of extra files.